



Fig. 3: Author's mesh antenna setup

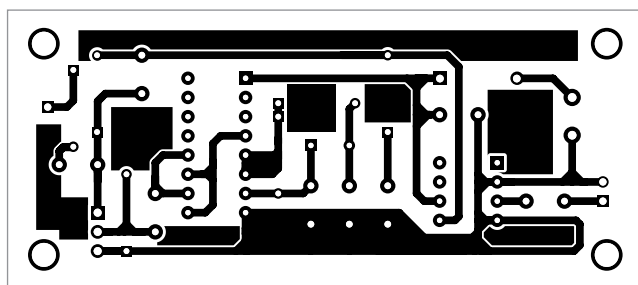


Fig. 4: Actual-size PCB layout of transmitter circuit

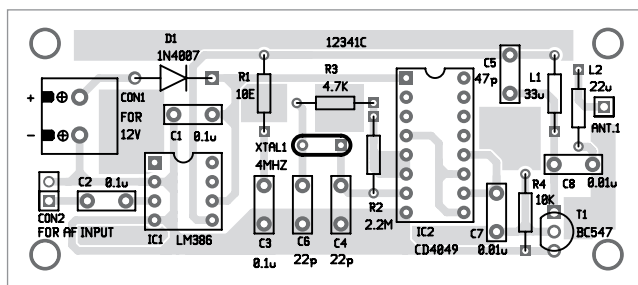


Fig. 5: Components layout of the PCB

modulated signal is obtained at its collector. The advantage of this method of modulation is that the output power can be easily increased by replacing the audio frequency amplifier with a higher power amplifier and replacing the transistor with one capable of handling more power.

The output of IC2 is a square wave and, according to Fourier analysis, a square wave consists of infinite series of sine waves of harmonics of the frequency of the square wave. The tuned

to get maximum output.

The output is coupled to the antenna by capacitor C8 and inductor L2. A wire of four to six metres length can be used as the antenna. Experimentally, the best results were obtained by using an insect-proof mesh (jali) in a window as antenna. The mesh was connected to the transmitter by using an ordinary 1.5 metre long wire. See Fig. 3.

The audio source used for testing was a 'Vire' brand USB player.

circuit comprising capacitor C5 and inductor L1 is designed to select only the fundamental frequency; thus the output is a sine wave of 4MHz frequency with the values of C5 and L1 selected for the circuit.

In practice, L1 has some internal capacitance also, so C5 may be replaced by a preset of 47pF as the resonance may occur at a capacitance lower than 47pF with the preset tuned

| PARTS LIST                                                    |                                            |
|---------------------------------------------------------------|--------------------------------------------|
| <i>Semiconductors:</i>                                        |                                            |
| IC1                                                           | - LM386 low-power amplifier                |
| IC2                                                           | - CD4049 quad hex inverter                 |
| D1                                                            | - 1N4007 rectifier diode                   |
| T1                                                            | - BC547 npn transistor                     |
| <i>Resistors (all 1/4-watt, <math>\pm 5\%</math> carbon):</i> |                                            |
| R1                                                            | - 10-ohm                                   |
| R2                                                            | - 2.2-mega-ohm                             |
| R3                                                            | - 4.7-kilo-ohm                             |
| R4                                                            | - 10-kilo-ohm                              |
| <i>Capacitors:</i>                                            |                                            |
| C1-C3                                                         | - 0.1 $\mu$ F ceramic disk                 |
| C4, C6                                                        | - 22pF ceramic disk                        |
| C5                                                            | - 47pF ceramic disk                        |
| C7, C8                                                        | - 0.01 $\mu$ F ceramic disk                |
| <i>Miscellaneous:</i>                                         |                                            |
| CON1                                                          | - 2-pin terminal connector                 |
| CON2                                                          | - 2-pin connector                          |
| XTAL1                                                         | - 4MHz crystal                             |
| L1                                                            | - 33 $\mu$ H inductor                      |
| L2                                                            | - 22 $\mu$ H inductor                      |
| ANT.1                                                         | - 6m wire/insect-proof mesh (jali) antenna |
|                                                               | - 12V battery or DC power supply           |
|                                                               | - 14-pin IC socket                         |
|                                                               | - 8-pin IC socket                          |

Excellent signals were received by using a Tecsun PL-390 radio receiver up to 500 metres away. If the audio source has a weak output, a 10 $\mu$ F capacitor can be connected between pins 1 and 8 of LM386. If the audio source has a strong output, it can be connected through a potentiometer to IC1.

The frequency of the oscillator can be changed by changing the crystal. An alternate method is to change L1 and C5 so that they form a tuned circuit that is tuned to first harmonic.

## Construction and testing

An actual-size PCB layout for the shortwave transmitter is shown in Fig. 4 and its components layout in Fig. 5.

After wiring, connect 12V DC power supply across CON1. Connect audio output from your MP3 player, mobile phone, or laptop to the input of your transmitter across CON2 connector. You are now ready to transmit and receive the signals in a shortwave radio up to 500 metres away. **EFY**

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